

Process Injection

Sub-techniques (12)

Adversaries may inject code into processes in order to evade process-based defenses as well as possibly elevate privileges. Process injection is a method of executing arbitrary code in the address space of a separate live process. Running code in the context of another process may allow access to the process's memory, system/network resources, and possibly elevated privileges. Execution via process injection may also evade detection from security products since the execution is masked under a legitimate process.

There are many different ways to inject code into a process, many of which abuse legitimate functionalities. These implementations exist for every major OS but are typically platform specific.

More sophisticated samples may perform multiple process injections to segment modules and further evade detection, utilizing named pipes or other inter-process communication (IPC) mechanisms as a communication channel.

ID: T1055

Sub-techniques: [T1055.001](#), [T1055.002](#), [T1055.003](#), [T1055.004](#), [T1055.005](#), [T1055.008](#), [T1055.009](#), [T1055.011](#), [T1055.012](#), [T1055.013](#), [T1055.014](#), [T1055.015](#)

-  Tactics: [Stealth](#), [Privilege Escalation](#)
-  Platforms: Linux, Windows, macOS

Contributors: Anastasios Pingios; Christiaan Beek, @ChristiaanBeek; Ryan Becwar

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack , Sandworm Team loaded BlackEnergy into svchost.exe, which then launched iexplore.exe for their C2. ^[1]
C0057	3CX Supply Chain Attack	During the 3CX Supply Chain Attack , AppleJeus 's VEILED SIGNAL uses process injection to inject the C2 communication module code in the first found process instance of Chrome, Firefox, or Edge web browsers. It also monitors the established named pipe and re-injects the C2 communication module if necessary. ^[2]
S0469	ABK	ABK has the ability to inject shellcode into svchost.exe. ^[3]
S0331	Agent Tesla	Agent Tesla can inject into known, vulnerable binaries on targeted hosts. ^[4]
S1074	ANDROMEDA	ANDROMEDA can inject into the <code>wuauclt.exe</code> process to perform C2 actions. ^[5]
G0050	APT32	APT32 malware has injected a Cobalt Strike beacon into Rundll32.exe. ^[6]
G0067	APT37	APT37 injects its malware variant, ROKRAT , into the cmd.exe process. ^[7]
G0082	APT38	APT38 has injected malicious payloads into the <code>explorer.exe</code> process. ^[8]
G0096	APT41	APT41 malware TIDYELF loaded the main WINTERLOVE component by injecting it into the iexplore.exe process. ^[9]
G1023	APT5	APT5 has used the CLEANPULSE utility to insert command line strings into a targeted process to alter its functionality. ^[10]
C0046	ArcaneDoor	ArcaneDoor included injecting code into the AAA and Crash Dump processes on infected Cisco ASA devices. ^[11]
S0438	Attor	Attor 's dispatcher can inject itself into running processes to gain higher privileges and to evade detection. ^[12]
S0347	AuditCred	AuditCred can inject code from files to other running processes. ^[13]
S0473	Avenger	Avenger has the ability to inject shellcode into svchost.exe. ^[3]
S0093	Backdoor.Oldrea	Backdoor.Oldrea injects itself into explorer.exe. ^{[14][15]}
S1081	BADHATCH	BADHATCH can inject itself into an existing explorer.exe process by using <code>RtlCreateUserThread</code> . ^{[16][17]}
S0534	Bazar	Bazar can inject code through calling <code>VirtualAllocExNuma</code> . ^[18]
S0470	BBK	BBK has the ability to inject shellcode into svchost.exe. ^[3]
G1043	BlackByte	BlackByte has injected Cobalt Strike into <code>wuauclt.exe</code> during intrusions. ^[19] BlackByte has injected ransomware into <code>svchost.exe</code> before encryption. ^[20]
S1181	BlackByte 2.0 Ransomware	BlackByte 2.0 Ransomware injects into a newly-created <code>svchost.exe</code> process prior to device encryption. ^[21]
S1039	Bumblebee	Bumblebee can inject code into multiple processes on infected endpoints. ^[22]
S0348	Cardinal RAT	Cardinal RAT injects into a newly spawned process created from a native Windows executable. ^[23]

ID	Name	Description
S0154	Cobalt Strike	Cobalt Strike can inject a variety of payloads into processes dynamically chosen by the adversary. ^{[27][28][29]}
S0614	CostaBricks	CostaBricks can inject a payload into the memory of a compromised host. ^[30]
C0029	Cutting Edge	During Cutting Edge , threat actors used malicious SparkGateway plugins to inject shared objects into web process memory on compromised Ivanti Secure Connect VPNs to enable deployment of backdoors. ^[31]
S0695	Donut	Donut includes a subproject <code>DonutTest</code> to inject shellcode into a target process. ^[32]
S9021	DOWNIISSA	DOWNIISSA can inject shellcode directly into process memory including WINWORD.exe and msixexec.exe. ^[33]
S1159	DUSTTRAP	DUSTTRAP compromises the <code>.text</code> section of a legitimate system DLL in <code>%windir%</code> to hold the contents of retrieved plug-ins. ^[34]
S0024	Dyre	Dyre has the ability to directly inject its code into the web browser process. ^[35]
S0554	Egregor	Egregor can inject its payload into iexplore.exe process. ^[36]
S0363	Empire	Empire contains multiple modules for injecting into processes, such as <code>Invoke-PSInject</code> . ^[37]
G0047	Gamaredon Group	Gamaredon Group has injected Remcos into explorer.exe. ^[38]
S0168	Gazer	Gazer injects its communication module into an Internet accessible process through which it performs C2. ^[39] ^[40]
S0032	gh0st RAT	gh0st RAT can inject malicious code into process created by the "Command_Create&Inject" function. ^[41]
S0561	GuLoader	GuLoader has the ability to inject shellcode into a donor processes that is started in a suspended state. GuLoader has previously used RegAsm as a donor process. ^[42]
S9023	HiddenFace	HiddenFace can inject code directly into legitimate applications. ^[43]
S0376	HOPLIGHT	HOPLIGHT has injected into running processes. ^[44]
S0040	HTRAN	HTRAN can inject into into running processes. ^[45]
S0398	HyperBro	HyperBro can run shellcode it injects into a newly created process. ^[46]
S0260	InvisiMole	InvisiMole can inject itself into another process to avoid detection including use of a technique called ListPlanting that customizes the sorting algorithm in a ListView structure. ^[47]
S0581	IronNetInjector	IronNetInjector can use an IronPython scripts to load a .NET injector to inject a payload into its own or a remote process. ^[48]
S0044	JHUHUGIT	JHUHUGIT performs code injection injecting its own functions to browser processes. ^{[49][50]}
S0201	JPIN	JPIN can inject content into lsass.exe to load a module. ^[51]
G0094	Kimsuky	Kimsuky has used Win7Elevate to inject malicious code into explorer.exe. ^[52]
S0681	Lizar	Lizar can migrate the loader into another process. ^[53]
S9020	LODEINFO	LODEINFO can inject shellcode into the memory of compromised hosts. ^{[54][55][56]}
		^[57]

ID	Name	Description
S0198	NETWIRE	NETWIRE can inject code into system processes including notepad.exe, svchost.exe, and vbc.exe. ^[62]
S1100	Ninja	Ninja has the ability to inject an agent module into a new process and arbitrary shellcode into running processes. ^{[63][64]}
S9025	NOOPLDR	NOOPLDR can inject decrypted payloads into processes including wuaucft.exe., rdrleakdiag.exe, and tabcal.exe. ^[65]
C0013	Operation Sharpshooter	During Operation Sharpshooter , threat actors leveraged embedded shellcode to inject a downloader into the memory of Word. ^[66]
C0014	Operation Wocao	During Operation Wocao , threat actors injected code into a selected process, which in turn launches a command as a child process of the original. ^[67]
S0664	Pandora	Pandora can start and inject code into a new <code>svchost</code> process. ^[68]
S1050	PcShare	The PcShare payload has been injected into the <code>logagent.exe</code> and <code>rdpclip.exe</code> processes. ^[69]
G0068	PLATINUM	PLATINUM has used various methods of process injection including hot patching. ^[51]
S0378	PoshC2	PoshC2 contains multiple modules for injecting into processes, such as <code>Invoke-PSInject</code> . ^[70]
S9019	PureCrypter	PureCrypter can inject its final stage into another process on the targeted system. ^[71]
S0650	QakBot	QakBot can inject itself into processes including explore.exe, lexplore.exe, Mobsync.exe., and wermgr.exe. ^[72] ^{[73][74][75][76]}
C0056	RedPenguin	During RedPenguin , UNC3886 exploited CVE-2025-21590 to enable malicious code injection into the memory of legitimate processes. ^{[77][78]}
S0332	Remcos	Remcos has a command to hide itself by injecting into another process. ^[79]
S0496	REvil	REvil can inject itself into running processes on a compromised host. ^[80]
S0240	ROKRAT	ROKRAT can use <code>VirtualAlloc</code> , <code>WriteProcessMemory</code> , and then <code>CreateRemoteThread</code> to execute shellcode within the address space of <code>Notepad.exe</code> . ^[81]
S0446	Ryuk	Ryuk has injected itself into remote processes to encrypt files using a combination of <code>VirtualAlloc</code> , <code>WriteProcessMemory</code> , and <code>CreateRemoteThread</code> . ^[82]
S0596	ShadowPad	ShadowPad has injected an install module into a newly created process. ^[83]
G0091	Silence	Silence has injected a DLL library containing a Trojan into the fwmain32.exe process. ^[84]
S0692	SILENTRINITY	SILENTRINITY can inject shellcode directly into Excel.exe or a specific process. ^[85]
S0633	Sliver	Sliver includes multiple methods to perform process injection to migrate the framework into other, potentially privileged processes on the victim machine. ^{[86][87][88][89]}
S0533	SLOTHFULMEDIA	SLOTHFULMEDIA can inject into running processes on a compromised host. ^[90]
S0226	Smoke Loader	Smoke Loader injects into the Internet Explorer process. ^[91]
S0380	StoneDrill	StoneDrill has relied on injecting its payload directly into the process memory of the victim's preferred ^[92]

ID	Name	Description
S0436	TSCookie	TSCookie has the ability to inject code into the svchost.exe, iexplorer.exe, explorer.exe, and default browser processes. ^[96]
G0010	Turla	Turla has also used PowerSploit 's <code>Invoke-ReflectivePEInjection.ps1</code> to reflectively load a PowerShell payload into a random process on the victim system. ^[97]
G1047	Velvet Ant	Velvet Ant initial execution included launching multiple <code>svchost</code> processes and injecting code into them. ^[98]
S0670	WarzoneRAT	WarzoneRAT has the ability to inject malicious DLLs into a specific process for privilege escalation. ^[99]
S0579	Waterbear	Waterbear can inject decrypted shellcode into the LanmanServer service. ^[100]
S0206	Wiarp	Wiarp creates a backdoor through which remote attackers can inject files into running processes. ^[101]
S0176	Wingbird	Wingbird performs multiple process injections to hijack system processes and execute malicious code. ^[102]
G0102	Wizard Spider	Wizard Spider has used process injection to execute payloads to escalate privileges. ^[103]
S1065	Woody RAT	Woody RAT can inject code into a targeted process by writing to the remote memory of an infected system and then create a remote thread. ^[104]

Mitigations

ID	Mitigation	Description
M1040	Behavior Prevention on Endpoint	Some endpoint security solutions can be configured to block some types of process injection based on common sequences of behavior that occur during the injection process. For example, on Windows 10, Attack Surface Reduction (ASR) rules may prevent Office applications from code injection. ^[105]
M1026	Privileged Account Management	Utilize Yama (ex: /proc/sys/kernel/yama/ptrace_scope) to mitigate ptrace based process injection by restricting the use of ptrace to privileged users only. Other mitigation controls involve the deployment of security kernel modules that provide advanced access control and process restrictions such as SELinux, grsecurity, and AppArmor.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0508	Behavioral Detection of Process Injection Across Platforms	AN1399	Detects process injection by correlating memory manipulation API calls (e.g., VirtualAllocEx, WriteProcessMemory), suspicious thread creation (e.g., CreateRemoteThread), and unusual DLL loads within another process's context.
		AN1400	Detects ptrace- or memfd-based process injection through audit logs capturing system calls (e.g., ptrace, mmap) targeting running processes along with suspicious file descriptors or memory writes.
		AN1401	Detects memory-based injection by monitoring <code>task_for_pid</code> , <code>mach_vm_write</code> , and dylib injection patterns through <code>DYLD_INSERT_LIBRARIES</code> or manual memory mapping.

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